

Q 20 - Validating trig identities

For each of these “identities,” determine whether it is *valid* or *invalid* for all θ .

If valid, derive it from more basic sine-cosine identities.

1. $\cos(2\theta) = \cos^2 \theta - \sin^2 \theta$

 Answer >

Valid. Fundamental double-angle form.

2. $\cos(2\theta) = 1 - 2 \sin^2 \theta$

 Answer >

Valid. Rearrangement of double-angle using $\cos^2 = 1 - \sin^2$.

3. $\cos(2\theta) = 2 \cos^2 \theta + 1$

 Answer >

Invalid. The correct form is $2 \cos^2 \theta - 1$.

4. $\tan(2\theta) = \frac{2 \tan \theta}{1 - \tan^2 \theta}$

 Answer >

Valid. Standard tangent double-angle formula.

5. $\tan(2\theta) = \frac{2 \tan \theta}{1 + \tan^2 \theta}$

 Answer >

Invalid. Wrong sign in the denominator.

6. $\tan^2 \theta = \sec^2 \theta - 1$

 **Answer** >

Valid. From $1 + \tan^2 \theta = \sec^2 \theta$.

7. $\csc^2 \theta = 1 - \cot^2 \theta$

 **Answer** >

Invalid. Should be $1 + \cot^2 \theta = \csc^2 \theta$.

8. $\sin^2 \theta = 1 + \cos^2 \theta$

 **Answer** >

Invalid. The correct form is $\sin^2 \theta + \cos^2 \theta = 1$.

9. $\sin(2\theta) = 2 \sin \theta \cos \theta$

 **Answer** >

Valid. Standard double-angle identity.

10. $\sin^2 \theta = \frac{1 + \cos(2\theta)}{2}$

 **Answer** >

Invalid. Should be $\sin^2 \theta = \frac{1 - \cos(2\theta)}{2}$.

11. $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$

≡ Answer >

Valid. Standard half-angle rewrite for $\cos^2 \theta$.

12. $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$

≡ Answer >

Invalid. Should be $\cos^2 \theta = \frac{1 + \cos(2\theta)}{2}$.

13. $\tan(\pi + \theta) = \tan \theta$

≡ Answer >

Valid. Tangent has period π .

14. $\tan(\pi - \theta) = \tan \theta$

≡ Answer >

Invalid. $\tan(\pi - \theta) = -\tan \theta$.

15. $\csc(-\theta) = -\csc \theta$

≡ Answer >

Valid. Sine is odd so cosecant is odd.

16. $\sec(-\theta) = -\sec \theta$

≡ Answer >

Invalid. Cosine is even so secant is even.

17. $\cot^2 \theta + 1 = \csc^2 \theta$

≡ Answer >

Valid. Pythagorean identity for cot / csc.

18. $1 + \cot^2 \theta = \sec^2 \theta$

≡ Answer >

Invalid. RHS should be $\csc^2 \theta$, not $\sec^2 \theta$.